

## Demonstration of Proposed Features

|               |               |
|---------------|---------------|
| Date          | 27 June 2026  |
| Team ID       |               |
| Project Name  | Rising Waters |
| Maximum Marks | 1 Mark        |

### Demonstration of Proposed Features

This document presents the features proposed during the planning phase of the **Rising Waters** project. It indicates whether each feature was successfully implemented, demonstrated, and evaluated during project development.

| S.No | Feature Name                 | Description                                                                         | Status      | Demonstrated | Remarks                                                      |
|------|------------------------------|-------------------------------------------------------------------------------------|-------------|--------------|--------------------------------------------------------------|
| 1    | User Input Interface         | Allows users to enter rainfall, temperature, humidity, and other weather parameters | Implemented | Yes          | Built using Flask and HTML forms                             |
| 2    | Data Preprocessing           | Cleans and prepares the dataset before prediction                                   | Implemented | Yes          | Missing values handled and data normalized                   |
| 3    | Machine Learning Prediction  | Predicts flood occurrence using trained ML models                                   | Implemented | Yes          | Random Forest, Decision Tree, KNN, and XGBoost models tested |
| 4    | Prediction Result Display    | Displays whether flood risk is high or low based on user inputs                     | Implemented | Yes          | Results shown instantly on the web page                      |
| 5    | Model Performance Evaluation | Compares the accuracy of different machine learning algorithms                      | Implemented | Yes          | Best-performing model selected for deployment                |
| 6    | Web Application Deployment   | Deploys the project using Flask and IBM Cloud                                       | Implemented | Yes          | Application accessible through browser                       |

### Feature Implementation Summary

| Item                            | Value |
|---------------------------------|-------|
| Total Features Proposed         | 6     |
| Total Features Implemented      | 6     |
| Total Features Demonstrated     | 6     |
| Overall Implementation Rate (%) | 100%  |